

Multiclass Classification of Breast Cancer Subtypes Using High-Dimensional RNA-Seq Gene Expression Data

George David Apostolidis^{1*} and Merlijn Vriend^{1*}

^{1*}Wageningen University & Research, Droevendaalsesteeg 4,
Wageningen, 6708 PB, Gelderland, The Netherlands.

*Corresponding author(s). E-mail(s): george.apostolidis@wur.nl;
merlijn.vriend@wur.nl;

1 Introduction

Since machine learning can analyze high-dimensional genomic data, it has become a crucial tool in bioinformatics. Tens of thousands of genes can have their expression levels measured simultaneously by modern RNA-sequencing technologies, resulting in datasets with far more features than samples. Although these high-dimensional environments present considerable analytical difficulties, they also present a chance to find biologically significant patterns which help in medical judgment.

The categorization of breast cancer subtypes based on gene expression profiles is a significant use of machine learning in this context. Since different subtypes of breast cancer are linked to unique genetic characteristics, prognoses, and treatment outcomes, it is essential to accurately identify these subtypes because the disease is heterogeneous. Therefore, more focused treatment approaches and possibly better patient outcomes can be made possible by early and accurate classification.

In this project, we use the relative expression levels of 20,603 genes to categorize five subtypes of breast cancer: invasive breast cancer, invasive ductal carcinoma, invasive lobular carcinoma, invasive mixed mucinous carcinoma, and invasive breast cancer. With a noticeable class imbalance, the task is a multiclass classification problem. The goal of this study is to systematically improve upon a basic baseline classifier by assessing various preprocessing techniques and machine learning models. The macro-averaged F_1 -score, which weighs each subtype equally to account for class imbalance,

is the main metric used to evaluate model performance. Our goal is to find a reliable classification method that performs well when applied to unseen data through systematic experimentation and model comparison.

2 Methods

The three primary components of our experimental workflow were model selection, hyperparameter tuning, and data preprocessing. We took that action in order to methodically improve the provided baseline classifier. Because of its modular design, the contribution of each component was identified and evaluated, as well as its individual and combined effects on classification performance.

2.1 Data Preprocessing

The dataset contains 1,763 training samples with expression levels for 20,603 genes, representing a high dimensional setting ($p \gg n$). Additionally, the distribution of classes is significantly unbalanced, with the majority class accounting for about 78% of samples. We defined multiple preprocessing versions to assess their impact on model performance. The first preprocessing version (PP1) employed mean imputation and feature standardization to deal with a limited number of missing values. Standardization was used to guarantee that each gene made an equal contribution to model training for regularized linear models, where the magnitude of the coefficients depends on scale. In subsequent preprocessing, we included supervised feature selection and dimensionality reduction (PP2). We employed SelectKBest with ANOVA F-statistics to retain the most discriminative genes, reducing dimensionality while maintaining the traits most strongly associated with class differences. The feature space was then compressed into lower-dimensional components using Principal Component Analysis (PCA) in the final preprocessing step (PP3). Scikit-learn pipelines were integrated with all preprocessing steps to prevent data leakage during cross-validation.

2.2 Method Selection

We looked into three linear classifiers: Ridge Classifier (RC), Linear Support Vector Machine (LSVM), and Logistic Regression (LR). The three linear classifiers that we evaluated were these models. For high-dimensional genomic data, linear models were chosen because simpler models often generalize better than highly flexible nonlinear alternatives.

To counteract the severe class imbalance and boost the impact of minority subtypes during training, all classifiers were set up with `class_weight="balanced"`. The macro-averaged F_1 -score was the main metric used to assess model performance. Unlike accuracy, which would be dominated by the majority class, this metric gives each class equal weight, guaranteeing that minority subtypes contribute equally to the overall evaluation.

Initial comparisons were performed without hyperparameter tuning to establish baseline performance under each preprocessing configuration. This allowed the precise effect of preprocessing on classification outcomes to be assessed.

2.3 Hyperparameter Tuning and Cross-Validation Strategy

We methodically adjusted the hyperparameters after identifying promising model-preprocessing combinations. Since the Linear Support Vector Machine demonstrated the best and most reliable baseline performance, we gave it particular attention.

The hyperparameters were optimized using `GridSearchCV` with `RepeatedStratifiedKFold` cross-validation. The repeated stratified approach, which maintains class proportions in each fold while minimizing variance in performance estimates, is especially important in unbalanced multiclass settings. By adjusting the regularization parameter C over a logarithmic scale spanning multiple orders of magnitude, both highly regularized and less regularized regimes were captured. The ideal value of $C \approx 0.18$ indicates that moderate regularization produced the best balance between bias and variance.

All hyperparameter tuning was carried out within the pipeline framework to ensure that preprocessing transformations were refitted independently within each cross-validation fold. This design avoids information leakage while providing a reasonable estimate of the model’s generalization performance.

2.4 Final Model Training and Submission

After model selection and tuning, the ideal pipeline configuration was retrained using the complete training dataset. The final model was then applied to the unseen test set to generate predictions to submit to the Kaggle competition platform. Final performance was evaluated using the macro-averaged F1-score on the held-out test set.

This structured and leakage-safe experimental framework enabled a systematic comparison of models and preprocessing methods while maintaining robustness and reproducibility in performance estimation.

3 Results

3.1 Overview of Model Performance

Using stratified cross-validation and the macro-averaged F_1 -score as the main metric, we assessed various combinations of methods of preprocessing and linear classifiers. Macro- F_1 is more appropriate than accuracy because the dataset represents a high-dimensional setting (20,603 genes and 1,763 samples) with a severe class imbalance (approximately 78% majority class).

Figure 1 illustrates the dataset’s extreme imbalance with a single dominant subtype. The use of balanced class weights and macro-averaged evaluation metrics is justified by this imbalance, which presents a serious problem for multiclass classification.

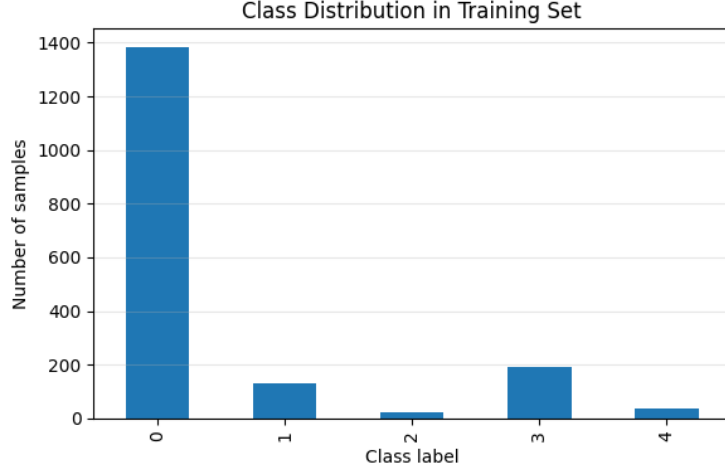


Fig. 1 Class distribution in the training set. Class 0 represents approximately 78% of all samples, while some subtypes contain fewer than 40 instances. This severe imbalance motivated the use of `class_weight="balanced"` and macro- F_1 as the primary evaluation metric.

The cross-validation macro- F_1 scores for the primary preprocessing-model combinations without hyperparameter tuning are summarized in Table 1.

Preprocessing	Model	CV Macro- F_1
PP1 (Impute + Scale)	ISVM	0.3599
PP3 (PCA)	ISVM	0.3348
PP3 (PCA)	RC	0.3300
PP2 (SelectKBest)	LR	0.3231
PP3 (PCA)	LR	0.3195
PP1 (Impute + Scale)	LR	0.3167
PP2 (SelectKBest)	ISVM	0.3134
PP2 (SelectKBest)	RC	0.3060
PP1 (Impute + Scale)	RC	0.2701

Table 1 Cross-validation macro- F_1 scores across preprocessing and model combinations.

Among the untuned models, the Linear Support Vector Machine with simple preprocessing (PP1) produced the best and most consistent cross-validation performance.

The methodical progress made during the experiment is shown in Figure 2. Switching from the baseline model to a properly regularized Linear SVM generated the biggest performance gain, underscoring the significance of model selection and tuning.

Performance was not consistently enhanced by advanced preprocessing techniques like feature selection and PCA.

Figure 3 shows how the cumulative explained variance gradually rises, necessitating a large number of principal components to preserve the majority of the variance. This

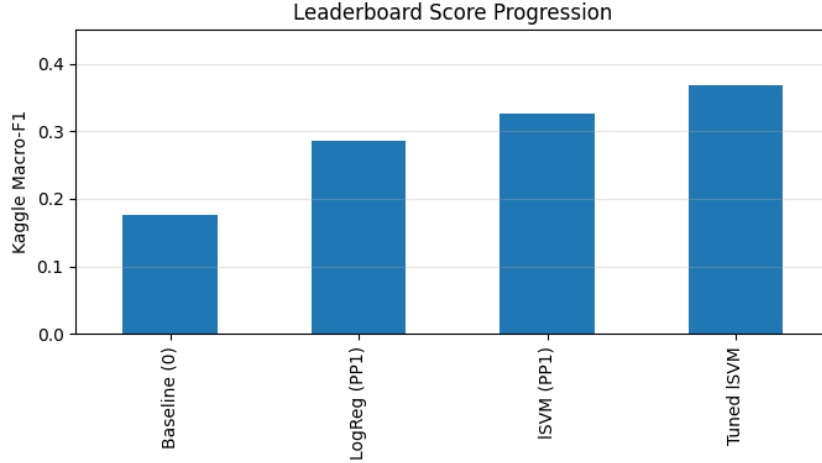


Fig. 2 Kaggle macro- F_1 progression across key experiments. Systematic improvements in preprocessing and model tuning resulted in a substantial performance increase compared to the baseline model.

could be the reason why classification performance was not consistently enhanced by PCA-based preprocessing.

3.2 Hyperparameter Tuning

In spite of its solid baseline performance, the Linear SVM was chosen for additional optimization. Using `GridSearchCV` with `RepeatedStratifiedKFold` cross-validation, hyperparameter tuning was carried out. Over a logarithmic scale covering several orders of magnitude, the regularization parameter C was adjusted.

The optimal value was found to be:

$$C \approx 0.18$$

with a cross-validation macro- F_1 score of:

$$\text{CV Macro-}F_1 = 0.3502.$$

This suggests that the optimal balance between bias and variance was achieved with moderate regularization.

3.3 Leaderboard Performance

Retraining the final tuned model on the entire training dataset and submitting it to the Kaggle platform allowed for the evaluation of generalization to unseen data.

Throughout the project, the Kaggle macro- F_1 scores' evolution is displayed in Table 2.

With a final Kaggle macro- F_1 score of **0.3686**, the tuned Linear SVM significantly outperformed the baseline model.

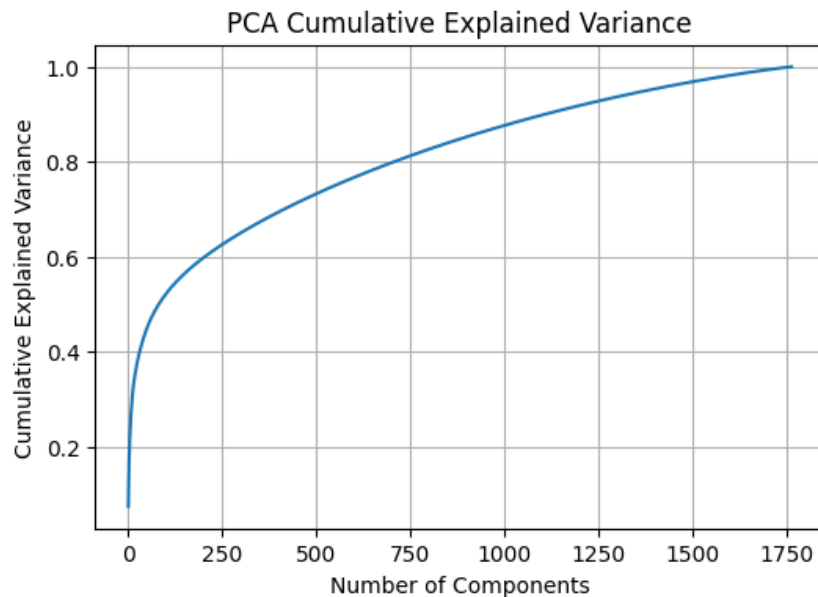


Fig. 3 Cumulative explained variance of PCA applied to standardized gene expression data. Variance is distributed across many components, indicating that aggressive dimensionality reduction may discard substantial information.

Model	Kaggle Macro- F_1
Baseline (predict class 0)	0.1760
Logistic Regression (PP1)	0.2857
Linear SVM (PP1)	0.3257
Tuned Linear SVM	0.3686

Table 2 Leaderboard performance progression.

It’s interesting to note that while some preprocessing techniques improved cross-validation scores, they had no effect on leaderboard performance. This implies that there may be some configurations where there is mild overfitting, leading to optimistic cross-validation estimates that did not generalize as well to the hidden test set.

3.4 Class-Level Behavior

The model identified classes 0–3 successfully, but it rarely predicted class 4, the smallest subtype, according to an analysis of predicted class distributions. Leaderboard performance did not improve further despite efforts to use modified class weights to increase the influence of rare classes. This implies a very small representation in the test set or a limited signal for this subtype.

Better balance between classes, rather than increases in overall accuracy, was probably the cause of improvements because macro- F_1 penalizes poor minority-class performance.

3.5 Conclusion from Results

The macro- F_1 score increased from 0.176 (baseline) to 0.3686 on unseen data through methodical experimentation and leakage-safe evaluation. Simple preprocessing and a moderately regularized Linear Support Vector Machine produced the best results. These results show that well-regularized linear models in addition to cautious cross-validation techniques can achieve competitive and robust performance in high-dimensional imbalanced genomic classification tasks.

4 Discussion and Conclusion

4.1 Discussion

The purpose of this study was to use high-dimensional gene expression data to better classify five subtypes of breast cancer. The outcomes show how competitive linear models are in this context, especially the Linear Support Vector Machine (LSVM). On unseen test data, the top-performing configuration obtained a macro- F_1 score of 0.3686, which is a significant improvement over the baseline model.

The method’s systematic and modular experimental design is one of its main advantages. We were able to clearly quantify each component’s contribution by keeping preprocessing, model selection, and hyperparameter tuning separate. Preprocessing steps were applied during cross-validation in a leakage-safe manner thanks to the use of scikit-learn pipelines. Additionally, in an unbalanced multiclass setting, the application of stratified and repeated cross-validation produced more stable performance estimates.

The findings suggest that in high-dimensional genomic classification, moderate regularization is crucial. Despite the large number of features relative to samples ($p \gg n$), overfitting was prevented by strong regularization. It’s interesting to note that more intricate preprocessing techniques like PCA and feature selection did not always result in better performance. They occasionally yielded better cross-validation results but did not generalize as well to the Kaggle test set. This draws attention to a possible drawback of depending only on cross-validation estimates, since they might still show optimistic bias.

Even with the advancements, there are still a number of restrictions. First, there are very few minority classes in the dataset, making it extremely unbalanced. The model hardly ever predicted the smallest subtype, even with `class_weight="balanced"`. This implies that either there are not enough training samples available for trustworthy pattern learning, or the signal for this subtype is weak. Secondly, only linear models were assessed. Even though linear models work well in high-dimensional environments, more sophisticated techniques like ensemble methods or elastic-net regularization may produce even better results. Lastly, the study only uses RNA-seq data, however, subtype discrimination may be enhanced by including additional biological or clinical features.

4.2 Conclusion

This project’s goal was to use gene expression data to enhance a basic baseline classifier for multiclass breast cancer subtype classification. We improved the macro- F_1 score on unseen test data from 0.176 to 0.3686 by using systematic hyperparameter tuning, leakage-safe preprocessing, and structured experimentation.

The results imply that robust cross-validation techniques in conjunction with well-regularized linear models can yield robust and generalizable performance in high-dimensional imbalanced genomic datasets. The current method offers a reliable and repeatable solution to the classification problem, though there may be room for improvement.

Overall, the research question can be answered in the affirmative: careful model selection and systematic preprocessing significantly enhance performance in multiclass breast cancer subtype classification when compared to a naive baseline.

5 ELSA Reflection

Several Ethical, Legal, and Social Aspects (ELSA) are brought up by the use of machine learning for the classification of breast cancer subtypes. Transparency, justice, privacy, accountability, and wider societal impact are some of these concerns.

Transparency is a major concern from an ethical standpoint. Despite being easier to understand than deep neural networks, linear support vector machines are still somewhat opaque decision-making systems. It’s possible that patients and clinicians don’t fully comprehend how gene expression patterns result in a particular subtype prediction. The ELSA framework emphasizes how a lack of transparency can erode trust and muddy accountability. Explainability is especially crucial in medical settings since treatment choices may be influenced by classification results.

Bias, justice, and fairness are also important. Certain cancer subtypes are under-represented in the highly unbalanced dataset used in this study. Minority classes run the risk of consistently performing worse as a result. Disparities still exist even after we applied `class_weight="balanced"` and assessed macro- F_1 to lessen imbalance. In a clinical context, disparities in subtype predictive performance may cause certain patient groups to receive treatment that is either inappropriate or delayed.

According to legal frameworks like the GDPR, genomic data is considered extremely sensitive personal data. Although this project made use of anonymized competition data, real-world implementation would necessitate stringent data governance, informed consent, safe storage, and adherence to data protection laws. Liability issues also come up: developers, medical professionals, and healthcare organizations may share accountability if a model incorrectly classifies a patient and this has an impact on treatment.

AI in healthcare may have a social impact on patient trust and professional autonomy. Under-reliance on algorithmic recommendations may limit the potential advantages of data-driven medicine, while over-reliance may diminish clinician decision-making autonomy. Therefore, it is crucial to make sure AI systems operate as tools for decision-making rather than as decision-makers.

All things considered, machine learning has a great deal of promise for enhancing cancer subtype classification and assisting with personalized medicine, however, its implementation must carefully consider accountability, transparency, fairness, and privacy. In addition to technical strength, ethical consideration and regulatory compliance are necessary for responsible integration into healthcare.

A Appendix

A.1 Experiment Log

PP	Model	HP	Acc (val)	F1 _{macro} (val)	Prec _{macro}	Rec _{macro}	Kaggle
1	LR	-	0.776	0.253	0.274	0.251	0.28566
1	ISVM	-	0.623	0.306	0.310	0.325	0.32574
1	RC	-	0.782	0.245	0.325	0.239	0.26998
1	LR	yes	—	0.3317 (CV)	—	—	—
1	ISVM	yes	—	0.3599 (CV)	—	—	—
1	RC	yes	—	0.2717 (CV)	—	—	—
2 (SelectKBest)	LR	yes	—	0.3899 (CV)	—	—	0.30792
2 (SelectKBest)	RC	yes	—	0.3749 (CV)	—	—	—
2 (SelectKBest)	ISVM	yes	—	0.3549 (CV)	—	—	—
3 (PCA)	LR	yes	—	0.3688 (CV)	—	—	—
3 (PCA)	ISVM	yes	—	0.3672 (CV)	—	—	—
3 (PCA)	RC	yes	—	0.3423 (CV)	—	—	—

Table 3 Experiment log summarizing preprocessing versions (PP), model types, hyperparameter tuning status, validation metrics, and Kaggle leaderboard performance.